

Betriebsanleitung

Raketenstart für Dummies

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R_2 and R_3 are in parallel. so they can be combined to a R_23

$$R_{23} = \frac{R_2 * R_3}{R_2 + R_3}$$

$$R_{23} = \frac{3300 * 5600}{3300 + 5600}$$

$$R_{23} = 2076.4044943820227$$

R_1 and R_23 are in series so

$$R_{23} + R_1 = R_{\text{total}}$$

$$R_{23} + R_1 = R_{\text{total}}$$

$$2076.4044943820227 + 1500 = R_{\text{total}}$$

$$R_{\text{total}} = 3576.4044943820227$$

$$V_{\text{total}} = V_1$$

$$V_{\text{total}} = 5$$

$$I_{\text{total}} = \frac{V_{\text{total}}}{R_{\text{total}}}$$

$$I_{\text{total}} = \frac{5}{3576.4044943820227}$$

$$I_{\text{total}} = 0.001398052152057807$$

The I through R_1 is V_total

$$I_{R1} = I_{\text{total}}$$

$$I_{R1} = 0.001398052152057807$$

$$V_{R1} = I_{R1} \times R_1$$

$$V_{R1} = 0.001398052152057807 \times 1500$$

$$V_{R1} = 2.0970782280867106$$

$$V_{R1} + V_{R23} = V_{\text{Total}}$$

$$V_{R23} = V_{\text{Total}} - V_{R1}$$

$$V_{R23} = 5 - 2.0970782280867106$$

$$V_{R23} = 2.9029217719132894$$

Because R_23 and R_3 and R_3 are in series;

$$V_{R23} = V_{R3} = V_{R2} = 2.9029217719132894$$

$$I_{R3} = \frac{V_{R3}}{R_3}$$

$$I_{R3} = \frac{2.9029217719132894}{5600}$$

$$I_{R3} = 0.0005183788878416588$$

And similarly for R2

$$I_{R2} = \frac{V_{R2}}{R_2}$$

$$I_{R2} = \frac{2.9029217719132894}{3300}$$

$$I_{R2} = 0.0008796732642161483$$

To summise.

$$I_{R1} = 0.001398052152057807$$

$$I_{R2} = 0.0008796732642161483$$

$$I_{R3} = 0.0005183788878416588$$

$$V_{R1} = 2.0970782280867106$$

$$V_{R2} = 2.9029217719132894$$

$$V_{R3} = 2.9029217719132894$$